

Norman Manigault

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EDUCATION

Arizona State University

Expected May 2028

BS in Computer Science

GPA: 3.73

Relevant Coursework: Data Structures and Algorithms, Digital Design and Fundamentals, Intro to Software Engineering

TECHNICAL SKILLS

Languages: Python, Typescript, Javascript, Java, C++, SQL

Libraries and Frameworks: Node.js, React, React Native, Next.js, FastAPI, Electron, Express, OpenCV, Pandas

Developer Tools: Docker, AWS, tRPC, Vite, Figma, Kubernetes, PostgreSQL, Git

WORK EXPERIENCE

The Luminosity Lab

Tempe, AZ

Student Researcher

Aug 2026 – Present

- Researched and architected a pipeline that translates drag-and-drop programs into compact commands for wired or Bluetooth execution on a Raspberry Pi Pico W robot, enabling accessible robotics coding for students
- Supporting backend services for authentication, classroom management, student progress tracking, payments, and AI-model integration through Amazon Bedrock.

Raytheon

Tucson, AZ

Software Engineer Intern

Jun 2026 – Aug 2026

- Built an internal tool automating the import of test limits and measured values into Excel, reducing manual data entry and improving reporting efficiency by **90%** using Python.
- Improved reliability of hardware test stations by supporting validation and verification of multiple electronic components, working with cross-functional teams, and utilizing C++ to enhance test processes.

Cita Marketplace

Remote

Software Engineer Intern

Jun 2025 – Aug 2025

- Implemented a feature for sellers to submit restaurant reservations, ensuring data validation and seamless integration with Next.js, TRPC, and SQL.
- Reduced user checkout time by **15%** by optimizing front-end user flows for reservations purchasing with Tailwind

PROJECTS

Quiver (Python, FastAPI, Electron, JavaScript, SQLite, SVN)

- Architected a cross-platform engineering PDM and project-management desktop application, exposing **54 REST endpoints** for team-scoped CAD vault synchronization, task dependencies, and Gantt planning.
- Built a SolidWorks metadata pipeline to extract custom properties and where-used data; secured multi-user workflows with bcrypt, JWT authentication, rotating refresh tokens, and 3-level role-based access control.

American Sign Language Detector (Python, Scikit-Learn, OpenCV, Pandas, Numpy)

- Engineered real-time ASL recognition system processing **36 characters** at 30 FPS by implementing a Random Forest Classifier trained on MediaPipe hand landmark features
- Scaled training dataset 10x through systematic data augmentation (rotational transforms, mirror flips, HSV color manipulation) improving model generalization and reducing overfitting.

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Management Leadership for Tomorrow

Washington, DC

Career Preparation Fellow

Dec 2025 – Present

- Accepted into a selective 18-month professional development program that accelerates the career growth of emerging leaders through structured coaching, mentorship, and targeted skill-building.
- Engage with leading partner organizations (e.g., LinkedIn, Bloomberg, and Deloitte) to gain industry exposure and insight into tech-talent pathways.

It Starts With Us

Washington, D.C

Co-Founder and Technical Director

Aug 2025 – Present

- Established volunteer organization engaging **2 community partners**, launching comprehensive youth programs that served **100+ students** through tutoring and mentorship initiatives
- Assisted the construction of **50+ volunteers** supporting two long-term partnerships, delivering over **200 hours** of service to local youth programs